Hybrid Quantum–Classical Neural Networks for Tuberculosis Detection from Chest X-ray Images

A Report Submitted
in Partial Fulfillment of the Requirements
for the Degree of
Bachelor of Technology
in
Computer Science & Engineering

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ALLAHABAD PRAYAGRAJ
November, 2026

Undertaking

We declare that the work presented in this report titled "*Hybrid Quantum–Classical Neural Networks for Tuberculosis Detection from Chest X-ray Images*", submitted to the Computer Science and Engineering Department, Motilal Nehru National Institute of Technology Allahabad, Prayagraj, for the award of the ***Bachelor of Technology*** degree in ***Computer Science & Engineering***, is our original work. We have not plagiarized or submitted the same work for the award of any other degree. In case this undertaking is found incorrect, we accept that our degree may be unconditionally withdrawn.

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Certificate

Certified that the work contained in the report titled "*Hybrid Quantum–Classical Neural Networks for Tuberculosis Detection from Chest X-ray Images*", by Saksham Tiwari (20233243), Prajjwal Singh (20233213), Madipelli Rithvik (20233176) and Sanchari Ray (20233250), has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

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Preface

This report describes a B.Tech. project that asks a narrow, testable question: when a small quantum circuit is inserted into an image classifier, does the classifier get better, and does it matter where the circuit is placed? We study the question on a real medical task, the detection of tuberculosis from chest radiographs, using circuits of four to ten qubits that are simulated on a laptop CPU.

The project combines the two halves of the subject in equal measure. On the quantum side it implements a variational quantum classifier, a quantum convolution ("quanvolution") layer, two data-encoding schemes and three entanglement patterns, and it measures the circuits themselves: expressibility, entangling capability, gradient scaling with qubit count and robustness to hardware noise. On the machine-learning side it builds a reproducible pipeline with fixed data splits, repeated seeds, matched classical controls, low-data experiments and paired statistical tests, so that every claim about the quantum components can be checked against a fair classical alternative.

The code, the configuration of every experiment and all raw results are included with this report so that each number in it can be regenerated.

Acknowledgements

We express our sincere gratitude to our project supervisor, Dr. Rajitha B., for the guidance, encouragement and critical feedback given throughout this project. The supervisor's insistence on fair comparisons and honest reporting shaped the way every experiment in this report was designed.

We thank the Department of Computer Science and Engineering, Motilal Nehru National Institute of Technology Allahabad, Prayagraj, for providing the academic environment and resources that made this work possible, and the faculty members whose courses in machine learning and quantum computing laid its foundation.

We are grateful to the authors of the TB Chest X-ray Database for making their data publicly available, and to the developers of PyTorch and PennyLane, whose open-source software was used for every experiment.

Finally, we thank our families and friends for their constant support.

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Abstract

Tuberculosis (TB) remains one of the leading infectious causes of death worldwide, and chest radiography is a standard first-line screening tool. This project investigates whether hybrid quantum–classical neural networks can improve automated TB detection, and compares two ways of combining the paradigms: a classical→quantum (C→Q) model, in which a convolutional neural network (CNN) feeds a four-qubit variational quantum classifier, and a quantum→classical (Q→C) model, in which a quantum convolution filter processes 2×2 pixel patches before a CNN. Both are evaluated on the TB Chest X-ray Database (3,500 normal and 700 TB images at 64×64) against a CNN baseline and four classical controls matched in structure and size, with circuits simulated in PennyLane.

On the full training set the C→Q hybrid reached $98.41 \pm 0.32\%$ accuracy and an F1-score of 0.952 ± 0.010 over three seeds, compared with $97.67 \pm 0.33\%$ and 0.930 ± 0.009 for the classical CNN and $97.72 \pm 0.40\%$ and 0.930 ± 0.012 for a parameter-matched classical head. With an ImageNet-pretrained ResNet18 backbone the quantum head reached $99.47 \pm 0.18\%$ accuracy. The improvements are consistent across seeds but not statistically significant in paired McNemar tests ($p \geq 0.09$). The Q→C hybrid matched the full CNN with about four times fewer parameters, but so did plain classical downsampling. Circuit analysis showed that the entangling layers approach Haar-random expressibility at depth two to three, that gradient variance decays roughly exponentially with qubit count, and that the trained hybrid keeps its accuracy with only ten measurement shots and under depolarising noise up to $p = 0.2$ per layer. At today's simulable scale, quantum layers are a viable, noise-robust component of a medical image classifier, but not yet a demonstrably superior one.

List of Abbreviations

AUC	Area Under the (ROC) Curve
BCE	Binary Cross-Entropy
BP	Barren Plateau
C→Q	Classical-to-Quantum hybrid (CNN features → quantum circuit)
CNN	Convolutional Neural Network
CNOT	Controlled-NOT gate
CXR	Chest X-ray (radiograph)
FN / FP	False Negative / False Positive
KL	Kullback–Leibler divergence
NISQ	Noisy Intermediate-Scale Quantum
PQC	Parameterised Quantum Circuit
Q→C	Quantum-to-Classical hybrid (quantum convolution → CNN)
QML	Quantum Machine Learning
ROC	Receiver Operating Characteristic
SEL	StronglyEntanglingLayers (circuit template)
TB	Tuberculosis
VQC	Variational Quantum Circuit / Classifier

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Chapter 1

Introduction

1.1 Motivation

Tuberculosis is caused by the bacterium *Mycobacterium tuberculosis* and most often affects the lungs. According to the World Health Organization, it was again the world's leading cause of death from a single infectious agent in 2023, with an estimated 10.8 million people falling ill and about 1.25 million deaths [54]. India carries the largest share of the global burden, which makes early and cheap detection a public-health priority in our own context.

Chest radiography is one of the main tools for TB screening. It is fast and inexpensive, but every image must be read by a trained radiologist, and in high-burden regions radiologists are scarce. The WHO therefore recommends computer-aided detection software as an option for screening adults [53], and an independent evaluation of five commercial products in Bangladesh found that all of them outperformed experienced radiologists at the operating points studied [40].

Such systems are built on convolutional neural networks (CNNs) [25]. Since a deep CNN won the ImageNet challenge [9; 22], deep learning has become the standard approach in computer vision [24] and in medical image analysis [26]; large public collections such as ChestX-ray8 [52] made it possible to train radiograph classifiers that approach radiologist-level performance [42]. For TB, ImageNet-pretrained CNNs reach accuracies in the high nineties on public datasets [41].

Quantum computing offers a different model of computation. Feynman observed that quantum systems are exponentially expensive to simulate classically [12]; Shor's

factoring algorithm [48] and Grover's search algorithm [14] showed provable speed-ups for specific problems; and in 2019 a 53-qubit processor performed a sampling task argued to be beyond classical supercomputers [2]. Present devices, however, belong to the *noisy intermediate-scale quantum* (NISQ) era [39]: they have few qubits and every gate adds noise. The dominant NISQ approach to machine learning is the *variational quantum circuit*, a short circuit with trainable rotation angles optimised by a classical computer like the weights of a neural network [3; 10; 31; 44]. The same principle underlies the variational quantum eigensolver [20; 38]; the family of methods is reviewed in [6; 46].

Because current devices are small, practical models are *hybrid*: a classical network handles the high-dimensional image and a small circuit is inserted at one point. In *classical-to-quantum* transfer learning a CNN compresses the image into a few features that a circuit classifies [27]; in a *quanvolutional* network a circuit is slid over the image like a convolution filter [17]. PennyLane [4] makes both kinds of model ordinary PyTorch [35] modules that can be trained by backpropagation through a simulator.

1.2 Problem Statement

Published hybrid models are often compared only with a classical model of a different size or structure, on a single run, so it is hard to tell whether the quantum circuit is responsible for the result. This project asks:

> *Does adding a small quantum circuit improve a CNN classifier for tuberculosis detection from chest X-rays, and does the answer depend on whether the circuit is placed after the classical feature extractor (C→Q) or before it (Q→C), once each hybrid is compared with a classical model of the same structure and size?*

1.3 Objectives and Contributions

The objectives were to build a reproducible pipeline for the TB Chest X-ray Database; to implement a classical CNN baseline, a C→Q hybrid and a Q→C

hybrid together with classical controls matched to each hybrid; to study the effect of circuit design, backbone and training-set size with repeated seeds and paired statistical tests; and to analyse the circuits themselves. The main contributions are:

1. A configurable PyTorch–PennyLane code base in which every experiment differs from its reference in exactly one setting.

2. A comparison of both hybrid placements with a baseline and four matched controls on five training-set sizes and three seeds (105 training runs), a 12-configuration circuit ablation, and a backbone $\times$ head study including pretrained ResNet18.

3. A quantitative analysis of the circuits: expressibility, entangling capability, barren-plateau scaling, and robustness to finite shots and depolarising noise.

4. An honest assessment: the C$\rightarrow$Q hybrid is consistently but not significantly better than its controls, and the Q$\rightarrow$C hybrid's parameter efficiency is reproduced by classical downsampling.

Chapter 2 reviews the background and related work, Chapter 3 describes the data, models and protocol, Chapter 4 presents and analyses the results, and Chapter 5 concludes.

Chapter 2

Related Work

2.1 Convolutional Neural Networks

A CNN applies small learned filters across the whole image [25]. A convolutional layer with C_{in} input channels, C_{out} output channels and a $k \times k$ kernel has $C_{\text{out}}(C_{\text{in}}k^2 + 1)$ parameters, independently of the image size, and recognises a feature wherever it appears. Each convolution is followed by a non-linearity, usually $\text{ReLU}(z) = \max(0, z)$ [32], and by pooling that reduces the resolution; fully connected layers then map the features to class scores [13]. For a binary task the output z is turned into a probability $\hat{p} = \sigma(z) = 1/(1 + e^{-z})$ and the network is trained by minimising the binary cross-entropy

$$\mathcal{L} = -\frac{1}{B}\sum_{b=1}^{B}\Big[y_b \log \hat{p}_b + (1 - y_b)\log(1 - \hat{p}_b)\Big] \tag{1}$$

over mini-batches of B labelled examples, with gradients from backpropagation and a stochastic optimiser such as Adam [21].

Deep architectures such as VGG [50] and ResNet [16], whose residual connections make very deep networks trainable, were developed on ImageNet [9]. Their early layers learn generic features that transfer well to other tasks [55], so fine-tuning a pretrained network is the standard approach when labelled medical data is scarce [26].

2.2 Deep Learning for Tuberculosis Detection

The Montgomery and Shenzhen datasets of Jaeger et al. [19], with a few hundred radiographs each, were the first widely used TB benchmarks. Lakhani and Sundaram [23] fine-tuned ImageNet-pretrained AlexNet and GoogLeNet on about a thousand radiographs and reached an area under the ROC curve of about 0.99 with an ensemble of the two. Pasa et al. [34] showed that a compact CNN with far fewer parameters reaches competitive accuracy, which supports the premise of this project that TB detection does not require a very large network.

Rahman et al. [41] assembled the dataset used here, the TB Chest X-ray Database, from several public sources (3,500 normal and 700 TB images). They compared nine pretrained CNNs with and without lung segmentation and reported accuracies of about 98–99%. Two points from their work recur in this report: the images come from different sources and acquisition conditions, and the best results use large networks on images of a few hundred pixels per side.

2.3 Quantum Computing Background

A qubit is a two-level system with basis states $|0\rangle$ and $|1\rangle$ and general state

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle = \cos\tfrac{\theta}{2}\,|0\rangle + e^{i\phi}\sin\tfrac{\theta}{2}\,|1\rangle\,, \qquad |\alpha|^2 + |\beta|^2 = 1, \tag{2}$$

a point on the Bloch sphere [33]. A register of n qubits has 2^n complex amplitudes, which is why quantum systems are hard to simulate and why they may represent functions that are expensive for classical models [5]. Computation uses unitary gates; the rotations used here are

$$R_Y(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}, \qquad R_Z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}, \tag{3}$$

$$\mathrm{Rot}(\phi, \theta, \omega) = R_Z(\omega) R_Y(\theta) R_Z(\phi), \tag{4}$$

together with $R_X(\theta) = e^{-i\theta X/2}$, where X, Y, Z are the Pauli matrices. The two-qubit CNOT gate flips its target when the control is $|1\rangle$; it creates *entanglement*, correlations that cannot be described qubit by qubit.

Measuring a qubit returns 0 or 1 with probabilities $|\alpha|^2$ and $|\beta|^2$, so quantum models output *expectation values* estimated from many repeated runs (*shots*):

$$\langle Z_k \rangle = \langle \psi | \, Z_k \, | \psi \rangle = P(0) - P(1) \in [-1, 1]. \tag{5}$$

With N shots the estimate has variance $(1 - \langle Z_k \rangle^2)/N$. Real devices are noisy and their state is a density matrix ρ; a common noise model is the depolarising channel

$$\mathcal{E}(\rho) = (1 - p)\, \rho + \frac{p}{3} \left(X \rho X + Y \rho Y + Z \rho Z \right), \tag{6}$$

which drives every $\langle Z_k \rangle$ towards zero.

2.4 Variational Quantum Circuits

A variational circuit applies a data encoding $S(\mathbf{x})$ and a trainable unitary $W(\boldsymbol{\theta})$ and measures,

$$f_k(\mathbf{x}; \boldsymbol{\theta}) = \langle 0 |^{\otimes n} \, S(\mathbf{x})^\dagger W(\boldsymbol{\theta})^\dagger \, Z_k \, W(\boldsymbol{\theta}) S(\mathbf{x}) \, | 0 \rangle^{\otimes n}, \tag{7}$$

and its outputs are fed to a classical loss. Schuld et al. [44] proposed "circuit-centric" classifiers built from layers of general rotations and CNOTs at increasing distances, available in PennyLane as `StronglyEntanglingLayers`; Mitarai et al. [31] showed that such circuits are universal function approximators when inputs are encoded non-linearly. Because the state cannot be read out on hardware, gradients are obtained with the *parameter-shift rule* [31; 43],

$$\frac{\partial f}{\partial \theta_j} = \frac{1}{2} \left[f\left(\theta_j + \tfrac{\pi}{2}\right) - f\left(\theta_j - \tfrac{\pi}{2}\right) \right], \tag{8}$$

which is exact and costs two circuit runs per parameter. In simulation, backpropagation through the simulator gives the same gradient more cheaply.

The encoding largely determines what the circuit can compute [45]. *Angle encoding* places one feature on each qubit, $\bigotimes_k R_Y(x_k) \, |0\rangle$; it needs n qubits for n features and one gate per qubit. *Amplitude encoding* writes a normalised vector of 2^n features into the amplitudes, $|\psi(\mathbf{x})\rangle = \|\mathbf{x}\|^{-1} \sum_i x_i \, |i\rangle$; it is exponentially compact but needs deep state preparation and discards the vector's length.

2.5 Hybrid Quantum–Classical Networks

Mari et al. [27] formalised classical-to-quantum transfer learning: a pretrained network is truncated to a short feature vector that a "dressed" circuit, with classical layers before and after it, classifies. Our C→Q model follows this design. Henderson et al. [17] introduced quanvolutional networks, in which a random, untrained circuit transforms each image patch; on MNIST these networks outperformed a plain CNN but performed similarly to a CNN with random non-linear classical filters, a finding that motivates the classical controls of our Q→C model. The quantum convolutional network of Cong et al. [8] is a different construction that acts on quantum input states with $\mathcal{O}(\log n)$ layers. Abbas et al. [1] argued from an effective-dimension measure that some quantum networks have higher capacity than comparable classical ones; capacity alone, however, does not guarantee better accuracy on a real task. Quantum kernel methods [15] and data re-uploading circuits [37] are further alternatives, discussed as future work.

2.6 Trainability and Circuit Quality

McClean et al. [28] showed that for sufficiently random circuits the gradient variance vanishes exponentially with the number of qubits, $\mathrm{Var}[\partial C/\partial\theta_j] \in \mathcal{O}(2^{-\alpha n})$ — a *barren plateau*. Cerezo et al. [7] showed that global costs can plateau even for shallow circuits, whereas local costs remain trainable. Sim et al. [49] proposed two descriptors of a circuit: *expressibility*, the Kullback–Leibler divergence between the distribution of fidelities F of pairs of its random output states and that of Haar-random states, $P_{\mathrm{Haar}}(F) = (N-1)(1-F)^{N-2}$ with $N = 2^n$ (lower is more expressive); and *entangling capability*, the mean Meyer–Wallach measure [30]

$$Q(|\psi\rangle) = 2\left(1 - \frac{1}{n}\sum_{k=1}^{n} \mathrm{Tr}\big(\rho_k^2\big)\right), \tag{9}$$

where ρ_k is the reduced state of qubit k ($Q = 0$ for product states). Holmes et al. [18] proved that the two properties conflict: the closer a circuit's states are to Haar-random, the smaller its gradients. On hardware, noise can additionally be reduced by error mitigation such as zero-noise extrapolation [51].

2.7 Research Gap

Hybrid models for medical images are usually reported with a single run and compared with a classical model that differs in size or structure, the two hybrid placements are rarely compared on the same data, and the circuits are seldom characterised beyond accuracy. This project evaluates both placements on one TB dataset against matched controls, with repeated seeds, several training-set sizes and paired significance tests, and combines these results with an analysis of the circuits' expressibility, trainability and noise robustness.

Chapter 3

Proposed Work

This chapter describes the data, the models and the evaluation protocol. All settings are the defaults in `configs/config.py` (Appendix A); each experiment changes exactly one of them.

3.1 Dataset and Preprocessing

We use the TB Chest X-ray Database of Rahman et al. [41]: 4,200 posterior–anterior radiographs, 3,500 *Normal* and 700 *Tuberculosis*, mostly at 512×512 pixels. Because the classes are imbalanced 5:1, a model that always answers "Normal" reaches 83.3% accuracy, so we report recall, F1 and ROC-AUC alongside accuracy and treat F1 as the main summary metric.

Each image is converted to grayscale, resized to 64×64 pixels and scaled to $[-1, 1]$; the resized images are cached once. No augmentation is used, so differences between models cannot come from augmentation randomness. The low resolution is a deliberate compromise that keeps more than 150 training runs, including 4.3 million simulated circuits for the quanvolution, feasible on a laptop CPU; we discuss it as a limitation.

The dataset has no official split, so we use a stratified 70/15/15 split with a fixed seed (Table 1). The split seed is separate from the training seed, so repeating an experiment with seeds 42, 43 and 44 changes only weight initialisation and batch order, and every model is scored on the same 630 test images. Low-data experiments draw a stratified 2%, 5%, 10% or 25% of the training split; the validation and test

sets are never reduced.

Table 1: Stratified split of the TB Chest X-ray Database used in all experiments.

Split	Normal	Tuberculosis	Total
Training	2,450	490	2,940
Validation	525	105	630
Test	525	105	630

3.2 Models

Every model maps an image through a *backbone* to a 32-dimensional embedding and through a *head* to the probability of TB. The hybrids differ from the classical baseline in one place only: the head (C→Q) or an input transform in front of the backbone (Q→C), as shown in Figure 1.

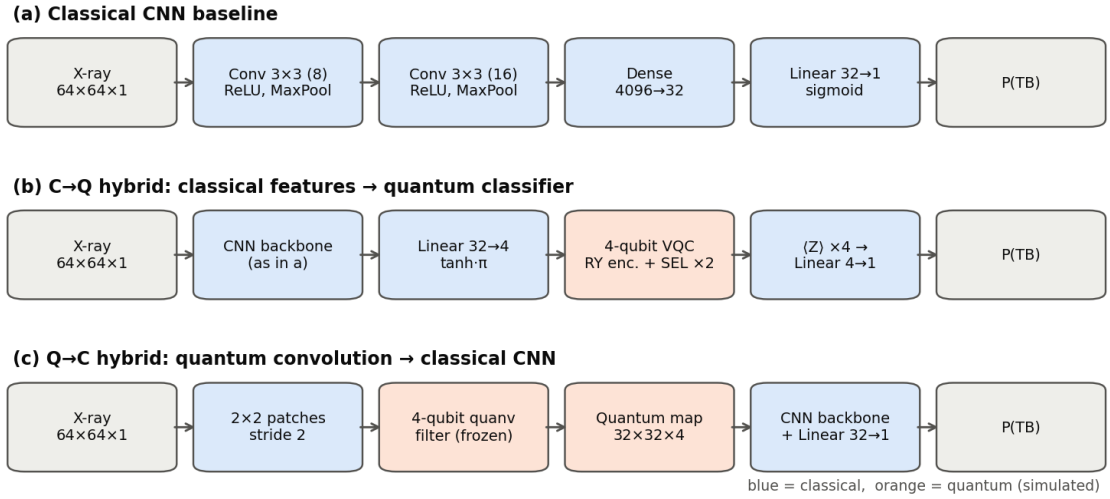


Figure 1: The three architectures: (a) classical CNN baseline, (b) classical-to-quantum hybrid, (c) quantum-to-classical hybrid. Blue blocks are classical, orange blocks are quantum.

3.2.1 Classical CNN Baseline

The backbone has two blocks of 3×3 convolution (8 and 16 filters, padding 1), ReLU [32] and 2×2 max pooling, followed by a dense projection of the $16 \times 16 \times 16$ feature map to 32 values (131,104 parameters). The classical head is a linear layer with a sigmoid (33 parameters). The model has 132,385 parameters in total; it is deliberately small, so that the quantum component is not hidden behind a very powerful classical network.

3.2.2 Classical-to-Quantum (C→Q) Hybrid

The C→Q hybrid replaces the linear head with a "dressed" variational quantum classifier [27]. A linear layer compresses the embedding to $n = 4$ values $\mathbf{u}$, which are scaled to $\mathbf{x} = \pi \tanh(\mathbf{u}) \in (-\pi, \pi)$ so that no two feature values map to the same rotation, and encoded as $\bigotimes_k R_Y(x_k) |0\rangle$. Two StronglyEntanglingLayers follow; each applies $\mathrm{Rot}(\phi, \theta, \omega)$ to every qubit and a ring of CNOTs of range 1 (first layer) and 2 (second layer), giving a weight tensor of shape $(L, n, 3)$, i.e. 24 parameters. The four expectation values $\langle Z_k \rangle$ (Eq. 5) pass through a linear layer and a sigmoid (Figure 2). The head has 161 parameters and the whole model 132,513, only 128 more than the baseline. The circuit is simulated with PennyLane's `default.qubit` simulator inside a `qml.qnn.TorchLayer`, so it trains like any PyTorch layer.

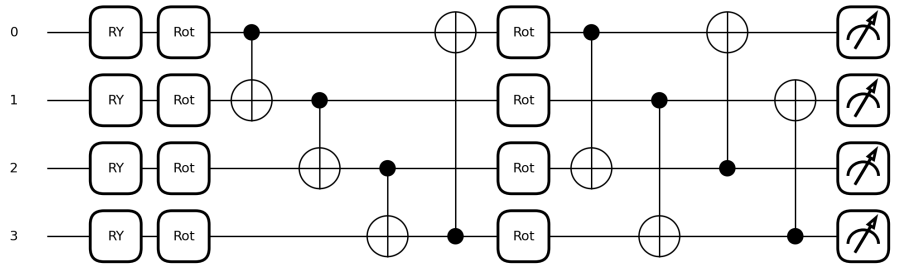


Figure 2: Quantum circuit of the C→Q head (4 qubits, depth 2): R_Y encoding, two StronglyEntanglingLayers, and measurement of $\langle Z \rangle$ on every qubit.

For the ablation the head can use depth $L \in \{1, 2, 3, 4\}$, $n \in \{2, 4, 6, 8\}$ qubits,

amplitude encoding (the compression layer then outputs 2^n values), BasicEntanglerLayers (one R_X per qubit and a nearest-neighbour CNOT ring), or no entanglement (the same rotations without CNOTs, equivalent to four independent one-qubit models).

3.2.3 Quantum-to-Classical (Q→C) Hybrid

The Q→C hybrid places a quantum circuit at the start of the network as a convolution filter [17]. The image is divided into 1,024 non-overlapping 2×2 patches; each pixel $p_k \in [0, 1]$ is encoded as $R_Y(\pi p_k)$, a fixed random circuit (one PennyLane `RandomLayers` layer, seed 0, Figure 3) is applied, and the four $\langle Z_k \rangle$ become four output channels. The resulting $4 \times 32 \times 32$ quantum feature map (Figure 4) is processed by the same CNN with four input channels. Because the input is half the size, the model has only 34,297 parameters. As in the original method the filter is random and frozen, so the transform is computed once for all 4,200 images (4.3 million four-qubit circuits) and cached.

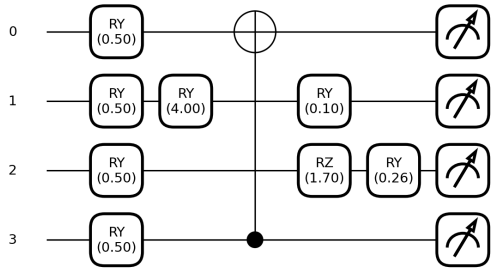


Figure 3: The quanvolution filter for one 2×2 patch (encoding angles set to 0.5 for display). Its gates are drawn once from a fixed seed and not trained.

3.2.4 Classical Control Models

A hybrid that beats a baseline does not prove that its circuit is useful, because it also differs in structure. Each hybrid is therefore paired with classical models that differ from it only in the quantum component (Table 2). The matched head

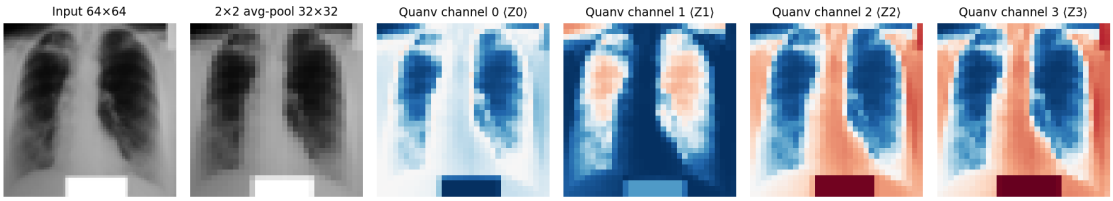


Figure 4: A TB radiograph at 64×64, its 2×2 average-pooled version, and the four channels of its quanvolution feature map. The rectangle at the bottom is a marker burned into the image, not anatomy.

mirrors the quantum head step for step, so any difference can only come from *what* the four-to-four transformation computes. The three Q→C controls separate three explanations: any fixed non-linear patch map, the downsampling itself, and a patch filter that is allowed to learn.

Table 2: Models in the main study and what each one controls for.

Model	Params	Purpose
Classical CNN	132,385	Baseline.
CNN + matched classical head	132,509	Control for C→Q: the circuit is replaced by a classical Linear$(4, 4)$ and tanh (157 head parameters versus 161).
C→Q hybrid	132,513	CNN + variational quantum head.
Q→C hybrid	34,297	Quanvolution + CNN.
Random patch + CNN	34,297	Control for Q→C: fixed random classical map $\tanh(W\pi\mathbf{p} + \mathbf{b})$ with the same input and output.
Avg-pool + CNN	34,081	Control for Q→C: plain 2×2 average pooling.
Learned 2×2 conv + CNN	34,317	Control for Q→C: trainable 2×2, stride-2 convolution with four channels and tanh.

3.2.5 Transfer Learning with ResNet18

To test the findings with a strong backbone, the small CNN is replaced by an ImageNet-pretrained ResNet18 [16]. The grayscale channel is repeated three times, the ImageNet classifier is replaced by a linear layer to the same 32-dimensional embedding, so either head can be attached, and the pretrained layers are fine-tuned with a ten times smaller learning rate (10^{-4}). The resulting models have about

11.19 million parameters.

3.3 Training and Evaluation Protocol

All models are trained identically: binary cross-entropy (Eq. 1), Adam [21] with learning rate 10^{-3}, batch size 32 and 15 epochs on a CPU, with no per-model tuning. In the main study every run keeps the weights of the epoch with the lowest validation loss, and runs on 2–10% of the data train for 40 epochs, since 15 epochs on 58 images are only about 30 gradient steps. Quantum gradients are computed by backpropagation through the simulator; on hardware the parameter-shift rule (Eq. 8) would be used, and we verify that the two agree.

An image is predicted as TB when its probability is at least 0.5. With TB as the positive class we report

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \qquad \text{Recall} = \frac{TP}{TP + FN}, \tag{10}$$

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}, \qquad \text{Precision} = \frac{TP}{TP + FP}, \tag{11}$$

and the ROC-AUC, the probability that a random TB image is scored above a random normal image [11]. Recall matters most clinically, since a missed TB case is more harmful than a false alarm. Multi-seed experiments report mean ± standard deviation over seeds 42, 43 and 44, and any "best" configuration is chosen on validation F1, never on the test set.

Because all models are scored on the same test images, two models are compared image by image with McNemar's exact test [29]. If b images are classified correctly by model A only and c by model B only, then under the null hypothesis of equal error rates $b \sim \text{Binomial}(b + c, \frac{1}{2})$. The test is run separately for each seed, and $p < 0.05$ is treated as significant.

The circuits are also characterised directly: expressibility and entangling capability of the trainable block (three entanglement patterns, depths 1–4, 5,000 random parameter pairs); the variance over 500 random initialisations of the gradient of a local cost $\langle Z_0 \rangle$ and a global cost $\langle Z^{\otimes n} \rangle$ for 2 to 10 qubits at depths 2 and 10; the trained C→Q model (seed 42) re-evaluated with 10 to 10,000 measurement shots

and with depolarising noise (Eq. 6) after every layer for p up to 0.2, with its weights fixed; and a comparison of parameter-shift and backpropagation gradients of the trained circuit.

Chapter 4

Experimental Setup and Results Analysis

4.1 Experimental Setup

All experiments ran on one laptop with an AMD Ryzen 7 6800H CPU (8 cores), 15.3 GB of memory and no GPU, under Windows 11 with Python 3.10.11, PyTorch 2.14.1, PennyLane 0.42.3, scikit-learn [36] 1.7.2 and NumPy 2.2.6. Circuits were simulated exactly with `default.qubit`, except in the noise experiments, which used finite shots and the `default.mixed` density-matrix simulator. Each experiment is a separate script that writes its configuration and metrics to `results/tb/`. Because the laptop was shared, training times varied between runs and are given as rough indications only. All results refer to the same 630 test images (525 normal, 105 TB).

In a first single-seed run (seed 42), replacing the 33-parameter linear head with the 161-parameter quantum head raised accuracy from 97.46% to 98.41% (16 errors against 10; missed TB cases 9 against 7) at about three times the training time. Since the two models differ in more than the circuit, the following experiments use repeated seeds and matched controls.

4.2 Circuit Design Ablation

The ablation varies one property of the C→Q circuit at a time, starting from 4 qubits, depth 2, angle encoding and strong entanglement (seed 42). Figure 5 shows

all twelve configurations on a common axis.

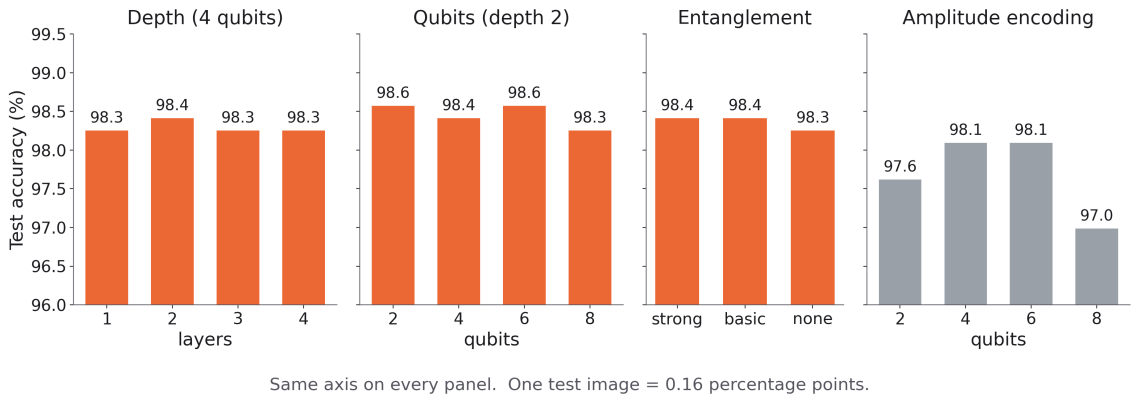


Figure 5: Test accuracy of all twelve circuit configurations of the C→Q hybrid (seed 42). All panels share one axis; one test image corresponds to 0.16 percentage points.

Depth changed test accuracy only between 98.25% and 98.41%, one test image, while training time grew roughly linearly with depth; the best validation F1 was obtained at depth 1. Strong and basic entanglement gave identical results (98.41%, F1 0.9515), and removing entanglement cost a single test image. With angle encoding, 2 to 8 qubits gave 98.25–98.57%. Amplitude encoding was worse in every case (96.98–98.10%) and worst at 8 qubits, where 256 amplitudes must be produced from a 32-dimensional embedding and normalisation discards the vector's length. Within the tested range the circuit design therefore barely affects accuracy: after the CNN has extracted its features, the four-to-one decision does not need the extra correlations that entanglement provides. Angle encoding is consistently preferable, and small circuits are as good as large ones.

4.3 Backbone and Head Comparison

The third experiment crosses two backbones with two heads, each with three seeds (Table 3).

With both backbones the quantum head gave a higher mean accuracy and F1 and a smaller standard deviation: 98.41±0.32% against 98.04±0.56% for the small CNN, and 99.47±0.18% against 99.21±0.57% for ResNet18. The backbone mattered about

Table 3: Backbone × head comparison: mean ± standard deviation over seeds 42, 43 and 44 on the fixed test set.

Model	Params	Accuracy (%)	Precision	Recall	F1	ROC-AUC	Time (s)
CNN + classical	132,385	98.04 ± 0.56	0.9326 ± 0.0286	0.9524 ± 0.0381	0.9418 ± 0.0172	0.9975 ± 0.0004	16
CNN + quantum	132,513	98.41 ± 0.32	0.9612 ± 0.0094	0.9429 ± 0.0165	0.9519 ± 0.0098	0.9980 ± 0.0014	46
ResNet18 + classical	11,192,961	99.21 ± 0.57	0.9875 ± 0.0216	0.9651 ± 0.0361	0.9758 ± 0.0178	0.9991 ± 0.0006	314
ResNet18 + quantum	11,193,089	99.47 ± 0.18	0.9936 ± 0.0056	0.9746 ± 0.0145	0.9839 ± 0.0057	0.9961 ± 0.0046	346

three times more than the head: ResNet18 raised accuracy by about one percentage point, at the cost of 85 times more parameters and about 20 times longer training. ROC-AUC exceeded 0.996 for every model, so all four rank images almost perfectly and the differences lie in where the 0.5 threshold falls. The head differences are comparable to the seed-to-seed variation and are tested for significance next.

4.4 Main Study: Hybrids versus Matched Controls

The main experiment trains the seven models of Table 2 with three seeds on five training-set sizes (105 runs). Table 4 gives the full-data results and Table 5 the paired tests.

Table 4: Main study on the full training set (2,940 images): mean ± std over three seeds. Head parameters count only the classifier head.

Model	Params	Head	Accuracy (%)	Recall	F1	ROC-AUC	Time (s)
Classical CNN	132,385	33	97.67 ± 0.33	0.9238 ± 0.0095	0.9298 ± 0.0094	0.9966 ± 0.0009	15
CNN + matched classical head	132,509	157	97.72 ± 0.40	0.9079 ± 0.0145	0.9301 ± 0.0119	0.9959 ± 0.0011	17
C→Q hybrid (CNN + quantum head)	132,513	161	98.41 ± 0.32	0.9429 ± 0.0165	0.9519 ± 0.0098	0.9980 ± 0.0014	43
Q→C hybrid (quanvolution + CNN)	34,297	33	97.88 ± 0.33	0.9270 ± 0.0145	0.9359 ± 0.0096	0.9954 ± 0.0017	5
Random patch + CNN (control)	34,297	33	90.53 ± 0.87	0.5206 ± 0.0715	0.6449 ± 0.0543	0.9336 ± 0.0035	4
Avg-pool + CNN (control)	34,081	33	98.15 ± 0.09	0.9556 ± 0.0110	0.9451 ± 0.0020	0.9956 ± 0.0008	4
Learned 2x2 conv + CNN (control)	34,317	33	98.31 ± 0.72	0.9524 ± 0.0000	0.9497 ± 0.0204	0.9956 ± 0.0033	7

Does the quantum head help? The C→Q hybrid was the best small-CNN model ($98.41 \pm 0.32\%$, F1 0.952 ± 0.010). Its matched classical control, with the same bottleneck, scored $97.72 \pm 0.40\%$ and 0.930 ± 0.012, almost exactly like the plain CNN ($97.67 \pm 0.33\%$, 0.930 ± 0.009), so the bottleneck alone does not explain the gain, and the hybrid was ahead for every seed. The McNemar tests, however, are not significant: $p = 0.092$, 0.146 and 1.000 against the matched head and 0.424,

Table 5: McNemar exact test between each hybrid and a classical comparator on the full-data runs. Each cell gives the p-value and, in brackets, the number of test images only the hybrid got right / only the comparator got right.

Hybrid	Compared with	Seed 42	Seed 43	Seed 44
C→Q hybrid (CNN + quantum head)	CNN + matched classical head	0.092 (10/3)	0.146 (9/3)	1.000 (7/7)
Q→C hybrid (quanvolution + CNN)	Random patch + CNN (control)	<0.001 (57/2)	<0.001 (49/7)	<0.001 (50/8)
Q→C hybrid (quanvolution + CNN)	Avg-pool + CNN (control)	1.000 (5/4)	0.344 (3/7)	0.791 (6/8)
Q→C hybrid (quanvolution + CNN)	Learned 2x2 conv + CNN (control)	1.000 (4/4)	1.000 (6/6)	0.057 (3/11)
C→Q hybrid (CNN + quantum head)	Classical CNN	0.424 (9/5)	0.227 (8/3)	0.267 (9/4)
Q→C hybrid (quanvolution + CNN)	Classical CNN	0.453 (5/2)	0.804 (7/9)	0.581 (8/5)

0.227 and 0.267 against the CNN. With error rates around 2%, two models disagree on only a handful of the 630 images, too few to rule out chance. The quantum head thus gives a small, consistent improvement of about 0.7 percentage points that this dataset cannot confirm statistically.

Does the quantum filter help? The Q→C hybrid reached $97.88 \pm 0.33\%$ with 34,297 parameters, indistinguishable from the 132,385-parameter CNN ($p = 0.453$, 0.804, 0.581). It was far better than the random classical patch map (90.53%, $p < 0.001$ for every seed), whose tanh saturates and turns patches into nearly binary values, while the quantum filter's cosine-shaped response keeps the detail. But 2×2 average pooling ($98.15 \pm 0.09\%$) and a learned 2×2 convolution ($98.31 \pm 0.72\%$) did as well as the quantum filter ($p \geq 0.057$). The four-fold parameter saving therefore comes from halving the resolution, not from the quantum computation, mirroring the finding of Henderson et al. [17].

Low-data regime. Table 6 gives F1 for smaller training sets. With 2% of the data (58 images), the Q→C hybrid and average pooling had the highest F1 (0.677 and 0.666), and the Q→C hybrid was by far the most stable model (standard deviation 0.020). The four-dimensional bottleneck heads were unreliable with little data: the matched classical head predicted "Normal" for every test image in all three seeds at 2% and in two of three seeds at 5% and 25%, and the C→Q hybrid reached only 0.341 ± 0.250 at 2%. This is a calibration failure rather than a failure to learn: at 2% the C→Q hybrid had the *highest* ROC-AUC of all models (0.927 against 0.911 for the CNN), so it ranks images well but its probabilities do not cross the 0.5

threshold in the right places. From 5% upwards the classical CNN had the highest or joint-highest F1, and only with the full data did the C→Q hybrid move ahead. The experiment therefore does not support a general claim that the hybrids learn more from fewer examples.

Table 6: Test F1 (mean ± std, three seeds) versus the fraction of the training split used. Validation and test sets are never reduced.

Model	2% (58)	5% (147)	10% (294)	25% (735)	100% (2940)
Classical CNN	0.601 ± 0.170	0.815 ± 0.027	0.880 ± 0.009	0.909 ± 0.008	0.930 ± 0.009
CNN + matched classical head	0.000 ± 0.000	0.263 ± 0.455	0.876 ± 0.010	0.309 ± 0.535	0.930 ± 0.012
C→Q hybrid (CNN + quantum head)	0.341 ± 0.250	0.787 ± 0.064	0.850 ± 0.077	0.871 ± 0.080	0.952 ± 0.010
Q→C hybrid (quanvolution + CNN)	0.677 ± 0.020	0.754 ± 0.093	0.834 ± 0.028	0.884 ± 0.015	0.936 ± 0.010
Random patch + CNN (control)	0.000 ± 0.000	0.416 ± 0.086	0.456 ± 0.065	0.490 ± 0.052	0.645 ± 0.054
Avg-pool + CNN (control)	0.666 ± 0.138	0.749 ± 0.099	0.862 ± 0.020	0.892 ± 0.011	0.945 ± 0.002
Learned 2x2 conv + CNN (control)	0.451 ± 0.076	0.705 ± 0.155	0.768 ± 0.092	0.857 ± 0.075	0.950 ± 0.020

4.5 Quantum Circuit Analysis

Figure 6 characterises the trainable block of the C→Q circuit without any data. With entangling gates, expressibility improves sharply from depth 1 to depth 2 (KL divergence 0.308 to 0.0083 for strong entanglement) and reaches 0.004–0.005 at depth 3, close to Haar-random. Without entanglement it stays at 0.19–0.30 whatever the depth, because product states can never cover the four-qubit state space. The entangling capability of the strong and basic patterns is 0.79–0.87, close to the Haar-random value $14/17 \approx 0.82$, and exactly zero without entanglement. Read together with the ablation, the circuits differ strongly as quantum objects — at depths 2 to 4 the no-entanglement circuit is 20 to 60 times less expressive — yet give almost the same accuracy: the task, as posed after a classical feature extractor, does not use the extra capacity.

Figure 7 shows how the gradient variance scales with the number of qubits. For depth 10 it falls from 0.150 at 2 qubits to 4.2×10^{-4} at 10 qubits for the local cost, and from 0.122 to 5.0×10^{-4} for the global cost — about a factor of two per added qubit, the 2^{-n} decay predicted by McClean et al. [28]. The depth-2 curves follow the same trend but are noisier, and we did not observe the advantage of local over global costs reported by Cerezo et al. [7], possibly because the long-range CNOTs spread

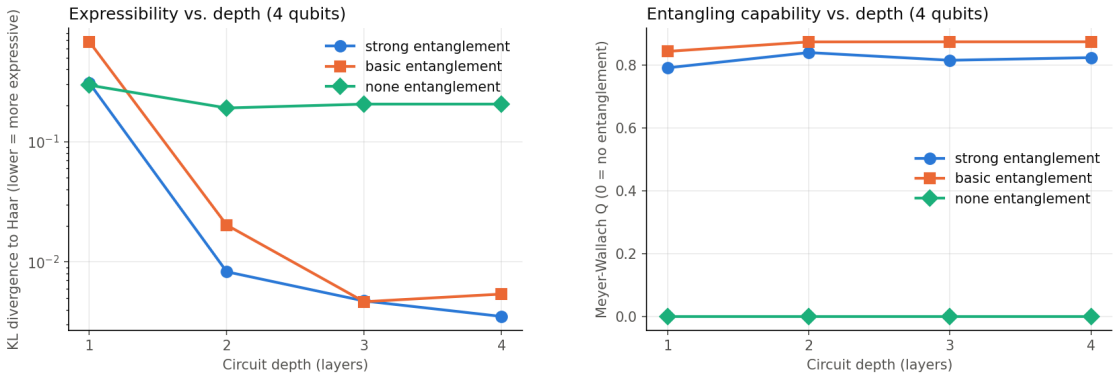


Figure 6: Expressibility (left, lower is more expressive) and entangling capability (right) of the three entanglement patterns versus depth, 4 qubits.

information across all qubits even at depth 2. At 4 qubits the variance is 60 to 300 times larger than at 10, which supports the choice of a small circuit, particularly for hardware, where each gradient must be estimated from finite shots.

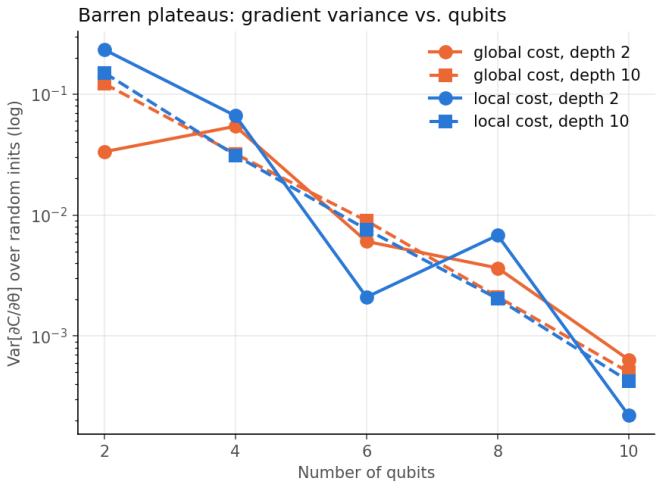


Figure 7: Variance of the cost gradient over 500 random initialisations versus number of qubits for local and global costs at depths 2 and 10.

4.6 Robustness to Shot Noise and Hardware Noise

Figure 8 shows the trained C→Q model (seed 42) evaluated as on an imperfect device. With only 10 shots per expectation value, which gives each $\langle Z_k \rangle$ a standard error of up to 0.32, accuracy was 98.25% against 98.41% for exact values, and with 50 to 10,000 shots it stayed within 98.25–98.57%. With depolarising noise after every layer it stayed within 98.41–98.57% up to $p = 0.2$, and the AUC was unchanged at 0.9987. Most test images lie far from the decision boundary, and depolarisation shrinks all four outputs by a similar factor (about 0.65–0.75 at $p = 0.1$), so the direction of the output vector and the ranking of images are preserved. Finally, parameter-shift and backpropagation gradients of the trained circuit agreed to 1.1×10^{-15}, so the model could be trained on hardware without any change. These results apply to small, shallow circuits; deeper circuits or less uniform noise would degrade more.

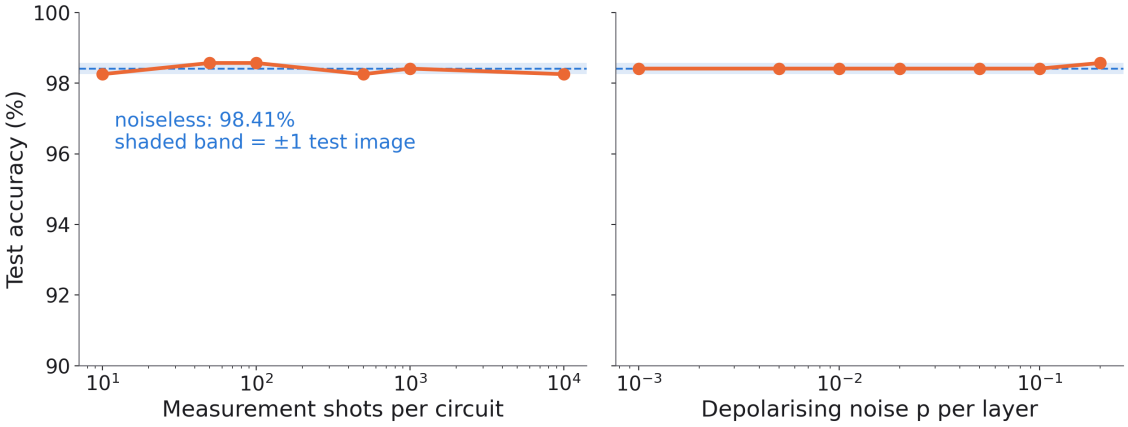


Figure 8: Test accuracy of the trained C→Q model versus measurement shots (left) and depolarising probability per layer (right). The shaded band is ±1 test image around the noiseless accuracy.

4.7 Error Analysis

For seed 42, only 22 of the 630 test images were misclassified by any of the four backbone × head models, and a single image by all four. The small-CNN models' errors were dominated by missed TB cases (9 of 16 for the classical head, 7 of 10 for the quantum head), the more harmful error in screening, while ResNet18 reduced the errors to one or two. Figure 9 shows the most confident mistakes of the CNN with a quantum head. They include TB cases with subtle changes that are hard to see at 64×64 pixels and a normal portable film with text markers, a reminder that models may partly learn differences between image sources rather than disease.

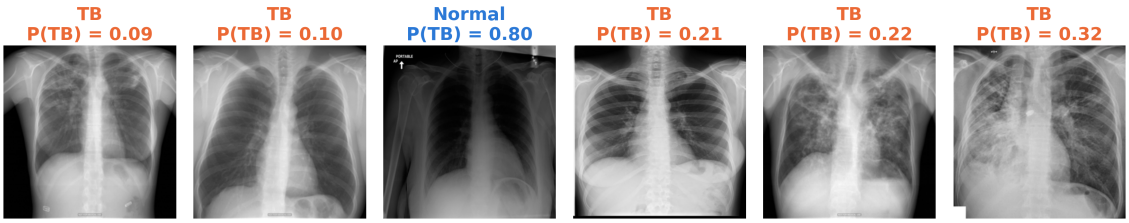


Figure 9: The six most confident mistakes of the CNN + quantum head model (seed 42), with the true class and the predicted probability of TB.

4.8 Discussion

Rahman et al. [41] reported 98–99% accuracy on the same dataset with large pre-trained CNNs on much larger images. Our classical CNN (97.67%), C→Q hybrid (98.41%) and ResNet18 with a quantum head (99.47%) fall in the same range, but the comparison is only indicative, since splits, preprocessing and resolution differ and a 630-image test set gives each accuracy an uncertainty of about ±1 percentage point.

Taken together, the experiments show that a quantum classifier head is a viable drop-in replacement for a classical one, with a small, consistent but not significant gain; that the quanvolution filter is a reasonable but not a special feature extractor, whose efficiency comes from downsampling; that circuit design changes expressibility by more than an order of magnitude but accuracy by at most a test image or

two; and that small hybrid circuits are trainable and robust to the noise of current hardware. What the evidence does *not* show is a quantum advantage: no hybrid was significantly better than its best classical counterpart, consistent with the expectation that small, classically simulable circuits cannot outperform classical models of similar size.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This project built and evaluated hybrid quantum–classical neural networks for detecting tuberculosis in chest radiographs, in two placements: a C→Q model in which a CNN's features are classified by a four-qubit variational circuit, and a Q→C model in which a four-qubit quanvolution filter transforms image patches before a CNN. Each hybrid was compared with classical controls of the same structure and size, with three seeds, five training-set sizes and paired McNemar tests, and the circuits were analysed directly. The main findings are:

1. The C→Q hybrid reached $98.41 \pm 0.32\%$ accuracy (F1 0.952 ± 0.010) against $97.72 \pm 0.40\%$ for a matched classical head and $97.67 \pm 0.33\%$ for the CNN; the gain was consistent across seeds but not significant ($p = 0.09$–1.00). With ResNet18 the quantum head reached $99.47 \pm 0.18\%$, and the backbone mattered about three times more than the head.

2. The Q→C hybrid matched the CNN with four times fewer parameters, but so did average pooling and a learned 2×2 convolution: the efficiency comes from downsampling, not from the quantum filter.

3. With 2% of the data the Q→C hybrid was the most stable model; bottleneck heads were unstable, although the quantum head still ranked images best (ROC-AUC 0.927).

4. Entanglement made the circuit 20–60 times more expressive but changed accuracy by at most one test image; angle encoding was consistently better than amplitude encoding.

5. Gradient variance halved with each added qubit, supporting small circuits, and the trained model kept its accuracy with 10 shots and depolarising noise up to $p = 0.2$, with exact parameter-shift gradients, so it is directly transferable to NISQ hardware.

Small quantum circuits can therefore be inserted into a medical image classifier without loss and, in the C→Q placement, with a small, consistent gain, but at the scale that can be simulated today they give no statistically demonstrable advantage over an equally sized classical component.

5.2 Limitations

All circuits were simulated with 2 to 10 qubits, and noise was modelled by an idealised depolarising channel. Images were reduced to 64×64 pixels, which may hide subtle findings. All results come from one public dataset whose normal and TB images were partly collected from different sources, so models may exploit acquisition differences such as markers and borders (Figure 4), and performance on images from another hospital could be lower. With 630 test images and three seeds, differences below about one percentage point cannot be confirmed. The quanvolution filter was random and frozen, hyperparameters were shared rather than tuned per model, and training times were measured on a shared laptop.

5.3 Future Work

The trained C→Q head should be run on a cloud quantum processor with the parameter-shift rule, readout error mitigation and zero-noise extrapolation [51]. The quanvolution filter could be made trainable and deeper so that its entanglement can contribute, and data re-uploading circuits [37] and quantum kernel methods [15] should be compared with the present heads. The low-data collapse of bottleneck

heads could be addressed by tuning the decision threshold on validation data or by a class-weighted loss. Training at 224×224 pixels and testing on independent collections such as the Montgomery and Shenzhen datasets [19] would measure generalisation across sources, and saliency methods such as Grad-CAM [47] would show whether the models attend to the lungs rather than to markers, as Rahman et al. [41] did for classical models. More seeds and cross-validation would provide the statistical power needed to confirm or reject the small advantage of the C$\rightarrow$Q head.

Appendix A

Configuration and Reproduction

Table 7 lists the fields of the configuration class in `configs/config.py` with their default values. Each experiment starts from these defaults and overrides only the field it studies.

With Python 3.10, the packages in `requirements.txt` and the dataset extracted into `data/tb_raw/`, all results are produced from the project root by

```
python -m src.train --model baseline
python -m src.train --model hybrid
python -m src.ablation
python -m src.compare
python -m src.hybrid_study
python -m src.quantum_analysis
python report/build_assets.py
```

Each script accepts `--subset` for a fast check on about 200 images. All runs are seeded (split seed 42; training seeds 42, 43, 44), and results are written to `results/tb/`.

Table 7: All configuration values (`configs/config.py`); experiments override only the swept field.

Field	Default value
seed	42
split_seed	42
dataset	tb
data_root	./data
image_size	64
batch_size	32
subset_train_size	200
subset_val_size	60
subset_test_size	60
tb_root	./data/tb_raw/TB_Chest_Radiography_Database
val_frac	0.15
test_frac	0.15
train_fraction	1.0
input_transform	none
quanv_layers	1
quanv_seed	0
backbone_name	small_cnn
conv1_channels	8
conv2_channels	16
embedding_dim	32
n_qubits	4
circuit_depth	2
encoding	angle
entanglement	strong
lr	0.001
pretrained_lr_scale	0.1
epochs	15
restore_best	False
results_dir	./results/tb

References

[1] Abbas, A., Sutter, D., Zoufal, C., Lucchi, A., Figalli, A., and Woerner, S. The power of quantum neural networks. *Nature Computational Science 1* (2021), 403–409.

[2] Arute, F., Arya, K., Babbush, R., Bacon, D., Bardin, J. C., Barends, R., et al. Quantum supremacy using a programmable superconducting processor. *Nature 574* (2019), 505–510.

[3] Benedetti, M., Lloyd, E., Sack, S., and Fiorentini, M. Parameterized quantum circuits as machine learning models. *Quantum Science and Technology 4*, 4 (2019), 043001.

[4] Bergholm, V., Izaac, J., Schuld, M., Gogolin, C., Killoran, N., et al. PennyLane: Automatic differentiation of hybrid quantum-classical computations. arXiv:1811.04968, 2018.

[5] Biamonte, J., Wittek, P., Pancotti, N., Rebentrost, P., Wiebe, N., and Lloyd, S. Quantum machine learning. *Nature 549* (2017), 195–202.

[6] Cerezo, M., Arrasmith, A., Babbush, R., Benjamin, S. C., Endo, S., Fujii, K., McClean, J. R., Mitarai, K., Yuan, X., Cincio, L., and Coles, P. J. Variational quantum algorithms. *Nature Reviews Physics 3* (2021), 625–644.

[7] Cerezo, M., Sone, A., Volkoff, T., Cincio, L., and Coles, P. J. Cost function dependent barren plateaus in shallow parametrized quantum circuits. *Nature Communications 12* (2021), 1791.

[8] Cong, I., Choi, S., and Lukin, M. D. Quantum convolutional neural networks. *Nature Physics 15* (2019), 1273–1278.

[9] Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., and Fei-Fei, L. ImageNet: A large-scale hierarchical image database. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (2009), pp. 248–255.

[10] Farhi, E., and Neven, H. Classification with quantum neural networks on near term processors. arXiv:1802.06002, 2018.

[11] Fawcett, T. An introduction to ROC analysis. *Pattern Recognition Letters 27*, 8 (2006), 861–874.

[12] Feynman, R. P. Simulating physics with computers. *International Journal of Theoretical Physics 21* (1982), 467–488.

[13] Goodfellow, I., Bengio, Y., and Courville, A. *Deep Learning*. MIT Press, Cambridge, MA, 2016.

[14] Grover, L. K. A fast quantum mechanical algorithm for database search. In *Proceedings of the 28th Annual ACM Symposium on Theory of Computing (STOC)* (1996), pp. 212–219.

[15] Havlíček, V., Córcoles, A. D., Temme, K., Harrow, A. W., Kandala, A., Chow, J. M., and Gambetta, J. M. Supervised learning with quantum-enhanced feature spaces. *Nature 567* (2019), 209–212.

[16] He, K., Zhang, X., Ren, S., and Sun, J. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (2016), pp. 770–778.

[17] Henderson, M., Shakya, S., Pradhan, S., and Cook, T. Quanvolutional neural networks: Powering image recognition with quantum circuits. *Quantum Machine Intelligence 2*, 1 (2020), 2.

[18] Holmes, Z., Sharma, K., Cerezo, M., and Coles, P. J. Connecting ansatz expressibility to gradient magnitudes and barren plateaus. *PRX Quantum 3* (2022), 010313.

[19] Jaeger, S., Candemir, S., Antani, S., Wáng, Y.-X. J., Lu, P.-X., and Thoma, G. Two public chest X-ray datasets for computer-aided screening of pulmonary diseases. *Quantitative Imaging in Medicine and Surgery 4*, 6 (2014), 475–477.

[20] Kandala, A., Mezzacapo, A., Temme, K., Takita, M., Brink, M., Chow, J. M., and Gambetta, J. M. Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets. *Nature 549* (2017), 242–246.

[21] Kingma, D. P., and Ba, J. Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)* (2015).

[22] Krizhevsky, A., Sutskever, I., and Hinton, G. E. ImageNet classification with deep convolutional neural networks. In *Advances in Neural Information Processing Systems 25 (NeurIPS)* (2012), pp. 1097–1105.

[23] Lakhani, P., and Sundaram, B. Deep learning at chest radiography: Automated classification of pulmonary tuberculosis by using convolutional neural networks. *Radiology 284*, 2 (2017), 574–582.

[24] LeCun, Y., Bengio, Y., and Hinton, G. Deep learning. *Nature 521* (2015), 436–444.

[25] LeCun, Y., Bottou, L., Bengio, Y., and Haffner, P. Gradient-based learning applied to document recognition. *Proceedings of the IEEE 86*, 11 (1998), 2278–2324.

[26] Litjens, G., Kooi, T., Bejnordi, B. E., Setio, A. A. A., Ciompi, F., Ghafoorian, M., van der Laak, J. A. W. M., van Ginneken, B., and Sánchez, C. I. A survey on deep learning in medical image analysis. *Medical Image Analysis 42* (2017), 60–88.

[27] Mari, A., Bromley, T. R., Izaac, J., Schuld, M., and Killoran, N. Transfer learning in hybrid classical-quantum neural networks. *Quantum 4* (2020), 340.

[28] McClean, J. R., Boixo, S., Smelyanskiy, V. N., Babbush, R., and Neven, H. Barren plateaus in quantum neural network training landscapes. *Nature Communications 9* (2018), 4812.

[29] McNemar, Q. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika 12*, 2 (1947), 153–157.

[30] Meyer, D. A., and Wallach, N. R. Global entanglement in multiparticle systems. *Journal of Mathematical Physics 43*, 9 (2002), 4273–4278.

[31] Mitarai, K., Negoro, M., Kitagawa, M., and Fujii, K. Quantum circuit learning. *Physical Review A 98* (2018), 032309.

[32] Nair, V., and Hinton, G. E. Rectified linear units improve restricted Boltzmann machines. In *Proceedings of the 27th International Conference on Machine Learning (ICML)* (2010), pp. 807–814.

[33] Nielsen, M. A., and Chuang, I. L. *Quantum Computation and Quantum Information*, 10th anniversary ed. Cambridge University Press, Cambridge, 2010.

[34] Pasa, F., Golkov, V., Pfeiffer, F., Cremers, D., and Pfeiffer, D. Efficient deep network architectures for fast chest X-ray tuberculosis screening and visualization. *Scientific Reports 9* (2019), 6268.

[35] Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., et al. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems 32 (NeurIPS)* (2019).

[36] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research 12* (2011), 2825–2830.

[37] Pérez-Salinas, A., Cervera-Lierta, A., Gil-Fuster, E., and Latorre, J. I. Data re-uploading for a universal quantum classifier. *Quantum 4* (2020), 226.

[38] Peruzzo, A., McClean, J., Shadbolt, P., Yung, M.-H., Zhou, X.-Q., Love, P. J., Aspuru-Guzik, A., and O'Brien, J. L. A variational eigenvalue solver on a photonic quantum processor. *Nature Communications 5* (2014), 4213.

[39] Preskill, J. Quantum computing in the NISQ era and beyond. *Quantum 2* (2018), 79.

[40] Qin, Z. Z., Ahmed, S., Sarker, M. S., Paul, K., Adel, A. S. S., Naheyan, T., et al. Tuberculosis detection from chest X-rays for triaging in a high tuberculosis-burden setting: An evaluation of five artificial intelligence algorithms. *The Lancet Digital Health 3*, 9 (2021), e543–e554.

[41] Rahman, T., Khandakar, A., Kadir, M. A., Islam, K. R., Islam, K. F., Mazhar, R., Hamid, T., Islam, M. T., Kashem, S., Mahbub, Z. B., Ayari, M. A., and Chowdhury, M. E. H. Reliable tuberculosis detection using chest X-ray with deep learning, segmentation and visualization. *IEEE Access 8* (2020), 191586–191601.

[42] Rajpurkar, P., Irvin, J., Zhu, K., Yang, B., Mehta, H., Duan, T., et al. CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning. arXiv:1711.05225, 2017.

[43] Schuld, M., Bergholm, V., Gogolin, C., Izaac, J., and Killoran, N. Evaluating analytic gradients on quantum hardware. *Physical Review A 99* (2019), 032331.

[44] Schuld, M., Bocharov, A., Svore, K. M., and Wiebe, N. Circuit-centric quantum classifiers. *Physical Review A 101* (2020), 032308.

[45] Schuld, M., and Killoran, N. Quantum machine learning in feature Hilbert spaces. *Physical Review Letters 122* (2019), 040504.

[46] Schuld, M., and Petruccione, F. *Supervised Learning with Quantum Computers*. Springer, Cham, 2018.

[47] Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., and Batra, D. Grad-CAM: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)* (2017), pp. 618–626.

[48] Shor, P. W. Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer. *SIAM Journal on Computing 26*, 5 (1997), 1484–1509.

[49] Sim, S., Johnson, P. D., and Aspuru-Guzik, A. Expressibility and entangling capability of parameterized quantum circuits for hybrid quantum-classical algorithms. *Advanced Quantum Technologies 2*, 12 (2019), 1900070.

[50] Simonyan, K., and Zisserman, A. Very deep convolutional networks for large-scale image recognition. In *International Conference on Learning Representations (ICLR)* (2015).

[51] Temme, K., Bravyi, S., and Gambetta, J. M. Error mitigation for short-depth quantum circuits. *Physical Review Letters 119* (2017), 180509.

[52] Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., and Summers, R. M. ChestX-ray8: Hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (2017), pp. 2097–2106.

[53] World Health Organization. WHO consolidated guidelines on tuberculosis. Module 2: Screening – systematic screening for tuberculosis disease. Tech. rep., World Health Organization, Geneva, 2021.

[54] World Health Organization. Global tuberculosis report 2024. Tech. rep., World Health Organization, Geneva, 2024.

[55] Yosinski, J., Clune, J., Bengio, Y., and Lipson, H. How transferable are features in deep neural networks? In *Advances in Neural Information Processing Systems 27 (NeurIPS)* (2014), pp. 3320–3328.